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AI Engineering Lead · Development leader, 10 years

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In one line

A development leader with 10 years of experience, turning research- and pilot-stage technology into products people actually use and pay for. I design and write the code myself, and have led teams from a 3-person dev unit to a 50-person research division.

Core strengths

01 From pilot technology to commercial product

- Led the monetization of [Alan](#), an LLM service piloted by a research team, from planning to launch as PO — Pro subscription launch (2025.07), first paid revenue (figures not disclosed)
- Built [Dr.Meta](#) and [Meta.CRO](#) on the output of [MX Studio](#), a 3D web component tool — in commercial operation at cancer centers nationwide

02 Designing AI systems hands-on, proving them in code

- Designed a code-generation pipeline combining rule-based modules with AI models and built its core (PoC) myself → officially launched with the team as [ProtoPie MCP](#) in October 2026
- Designed and built [Decision Graph](#), an LLM pipeline that extracts decisions, on my own — extracted 80%+ of the human-labeled decisions in a golden set

03 Deciding by measurement, not gut feel

- Model choices made by testing — rejected a switch to a lighter model that passed only 17 of 24 test cases, and restructured the calls for 74% fewer calls and 25% lower cost
- Set development priorities from traffic flows and per-feature usage in Google Analytics and Amplitude

04 Leadership sized to the team

- Built and led teams from a 3-person dev unit to a 50-person R&D division — as division head, acted as technical lead for a year while the CTO role was vacant; 3 group-wide awards
- Running an agile team of 4–5 AI engineers and app developers — build small, validate fast

Experience

Studio XID (ProtoPie) · Head of Product Dev Team

2026.01 - Present

Product Dev Team

Mission Take design-to-code AI from PoC to official launch — completed with the ProtoPie MCP launch (2026.10)

LLM pipeline for decision extraction (Slack-Notion → decision records with evidence)

2026.08 - 2026.10

Design & development (solo) · New product PoC: [Decision Graph](#)

- Turned PO requirements into a technical spec, then designed and built (solo) the LLM pipeline PoC that extracts who decided what and why from Slack and Notion, with source evidence
- Validated PoC results against a golden set where people manually marked the correct decisions on one real project — extracted 80%+ of the human-labeled decisions
- Verified and reviewed the feared technical risks firsthand through the PoC (judging quality against human-made ground truth, one-way records, preventing duplicate entries on re-runs)

- Chose the two-tier inference structure (first-pass classification: lightweight model; extraction and validation: high-capability model) by comparing options on the same tests (all-lightweight rejected after passing only 17 of 24)
- Logged time and cost per request to find bottlenecks and improved the call structure — on the same data, 74% fewer calls (1,289 → 336) and 25% lower cost (\$10.2 → \$7.6); re-running only the changed parts: \$0.7
- Checked quality on every change with 24 test cases with known correct answers (with a cap on evaluation cost)

AI-based code generation system (PoC → productization)

2026.01 - 2026.10

Head of Product Dev Team · Product: [ProtoPie MCP](#) (Dev View · Code MCP) official launch

- [Hands-on] Designed the code generation pipeline combining rule-based modules with AI model stages, and developed the core pipeline (PoC)
- [Hands-on] Designed training for auxiliary models that understand images and layout (structure), and built training data
- [Hands-on] Designed the product integration architecture for plugging the code generation pipeline into the existing product (ProtoPie)
- [Team lead] Ran an agile team of 4–5 AI engineers and app developers — model training by the AI engineers, integration development with the team
- [Team lead] Designed the MLOps pipeline from data cleaning → training → deployment
- [Team lead] Expanded to AOS, iOS and Web cross-platform; official launch as ProtoPie MCP in October 2026 (Dev View · Code MCP, code generation for 7 frameworks)

ESTsoft · Product Owner

2025.04 - 2026.01

AI Agent Lab / Alan Development Team / PO

Mission Turn the research team's pilot service Alan into a paid commercial product — completed with the Pro subscription launch (2025.07) and first paid revenue

AI search engine service (LLM-based agentic AI) Advancing 'Alan' (ESTsoft)

2025.04 - 2026.01

Product Owner · Product: Alan

- Led a service in trial operation at the research organization through commercial launch and monetization
- Led the development team for an LLM-based agentic AI app service, working as Product Owner
- Led a team made up of AI researchers and app developers (FE, BE)
- Operated and managed separate Dev, QA, Stage and Release servers
- Architected the Azure, Fast API, React and LangGraph structure
- Ran quality testing and collaborated with other in-house development, quality and infrastructure teams (QA, authentication, payments, infrastructure, etc.)
- Handled feature planning, scheduling and risk management for the LLM-based AI app service
- Built in-house image search, YouTube search & summarization and report generation features plus specialized agents
- Led planning of the slide generation feature — benchmarked Genspark and other services, defined user scenarios and the generation flow
- Designed the slide generation agent — staged orchestration (intent → scope and outline → research → template, style and fonts → HTML rendering → revision), with human-in-the-loop review of intermediate results
- Planned the monetization of an AI service — modeled margin from inference cost per call and set the Pro plan's price and usage limits from competitor pricing and expected paid conversion
- Led the paid launch as PO — built plan entitlements and payment, cancellation and refund flows, revised the terms and paid-conversion notices, and launched with marketing and business teams (2025.07, [announcement](#))

- Deployed the deep research and slide generation services, and developed an internal back-office operations tool
- Pro subscription brought the service's first paid revenue (figures not disclosed); building on it, planned an enterprise product (later the basis for institutional contract revenue)
- In commercial operation (myalan.ai)

Tmax Group · 8 yrs 3 mos

2017.01 - 2025.04

TmaxMetaAI · Project Leader, Development Team

2022.07 - 2025.04

Meta AI Research Division / Researcher (Head of Office)

Software for building 3D web components Development of 'MX studio' and productization based on it 2022.07 - 2025.02

Project Leader · Product: MX studio

- Led the MX studio engine team and a task force (planning, design, development, QA)
- Led design/development of the physics engine, real-time rendering & post-processing, 3D events/actions and a node-based visual code system
- Productized Dr.Meta and Meta.CRO on its output; ran an external beta test

Metaverse platform used by cancer centers nationwide Development of 'Dr.Meta' (Korea Smart Healthcare Association) 2024.06 - 2025.01

Project Leader · Product: Dr.Meta

- Led design/development of the React web app ↔ Unity app integration and admin/permission features
- In commercial operation at cancer centers nationwide (web.drmeta.kr)

3D-based virtual simulation training platform for clinical trials Development of 'Meta.CRO' (Korea Smart Healthcare Association) 2023.06 - 2025.01

Project Leader · Product: Meta.CRO

- Designed and released a 3D render engine library (private npm); led a development/QA task force
- In commercial operation at cancer centers nationwide

Online marketplace platform for contact lenses Development of 'NaenunN' (Pixelro) 2024.03 - 2024.09

Project Leader · Product: NaenunN

- The first online contact-lens purchase & delivery service, opened under a regulatory sandbox exemption — led the React Native cross-platform app; in commercial operation

GIS-based Samsung data center information system Development of 'GIS S-DCIS' (Samsung C&T) 2023.12 - 2024.08

Project Leader · Product: GIS S-DCIS

- GIS system on GeoServer · PostGIS · OpenLayers, built for a closed-network on-premise environment — co-developed and delivered with Yeonwoo Technology (B2B)

Real-time multi-sync server Research on 'MVS: Metaverse Server' (internal project) 2023.01 - 2024.09

Project Leader · Product: MVS

- Researched and developed a C++ real-time multi-user sync server engine — applied as the multiplayer server for Dr.Meta's medical staff conferences

TmaxAI · Project Leader	2020.07 - 2022.06
Meta Vision Research Division / Researcher (Head of Office, concurrently Division Head)	
- R&D Division Head (concurrently Head of Office) — led an R&D organization of up to 50 - Acted as technical lead for about a year while the CTO role was vacant	
Metaverse education platform Development of 'Mint'	2021.07 - 2022.06
Project Leader · Project: Mint	
Planned and built a metaverse education platform on Phaser · WebRTC · WebGL — delivered to Kongju National University	
Advanced technology research Deep Learning, 3DMM, SLAM	2020.07 - 2021.06
Project Leader · Research: Digital human and metaverse space research	
Early research on deep-learning digital humans (TTS) and SLAM-based 3D space generation — demoed at Tmax Day 2020	
TmaxA&C · Product Engineer	2017.01 - 2020.07
Graphics Research Division / Researcher (Team Lead)	
Collaboration solution Development of 'WAPL'	2019.01 - 2020.06
Product Engineer · Product: WAPL	
Frontend development of the collaboration solution's calendar and messenger — commercialized by TmaxWAPL	
Linux-based product Development of 'TmaxOS'	2017.06 - 2018.12
Product Engineer · Product: TmaxOS	
C++ development for a Linux-based OS — built and refactored the settings screen and video player	
Integrated system interface solution software Structure analysis of 'Anylink'	2017.02 - 2017.06
Structure analysis & refactoring design · Product: Anylink	
Structure analysis and refactoring design of an integrated interface solution — identified improvements with the engine research team; a TmaxSoft commercial product	

Skills

AI · LLM	Python · LLM API · LangGraph · MCP · RAG · Computer Vision
Backend · Infra	Node.js · FastAPI · Java · Spring · C/C++ · Docker · Jenkins · AWS · Azure
Frontend · Cross-platform	TypeScript · JavaScript · React · React Native · Three.js · Unity
Collaboration · Analytics	Git · Jira · Notion · Figma · Google Analytics · Amplitude
Platforms	Windows · Linux · iOS · Android

Education

KAIST, Culture Technology · M.S. in Culture Technology (Master's degree)	2013.09 - 2016.02 (Graduated)
3.5 / 4.3 · Human-computer interaction, computational design, game studies, IoT	
Ajou University (College of Information Technology), Digital Media · B.S. in Digital Media (Bachelor's degree)	2007.03 - 2013.08 (Graduated)
3.76 / 4.5 · Computer programming, computer architecture, object-oriented programming, information design	

Awards & activities

Award

- 2023.01 **Annual Technology Innovation Award** — Received the outstanding employee award titled 'Annual Technology Innovation Award' at TmaxMetaAI (Given once a year to about 10 people out of all employees across the group)
- 2022.06 **Super Leader Award** — Received the excellence award titled 'Super Leader' at TmaxAI (Given twice a year to about 5 people out of all employees across the group)
- 2022.01 **Maestro Award** — Received the top award titled 'Maestro' at TmaxAI (Given once a year to about 3 people out of all employees across the group)

Paper & research

- 2016.02 **Master's thesis**
Title: A Study on Mobile Game Design Strategies that Enhance Enjoyment and Self-Efficacy to Promote Everyday Physical Activity
- 2016.01.29 **Conference paper 2**
Title: A Study on Mobile Game Design Using Gamification and Symbolization Strategies to Promote Physical Activity in Everyday Life
Organizer: HCI Society of Korea · Authorship: First author · Pages: Vol.2016 No.1 [2016] p170-177 (Korean)
- 2016.01.28 **Conference paper 1**
Title: Gamification Can Affect the Sustainability of The Wearable Device for Physical Activity? : User Case Study
Organizer: HCI Society of Korea · Authorship: First author · Pages: Vol.2016 No.1 [2016] p451-455 (English)
- 2014.11 **Journal paper**
Title: An Exploratory Study of Factors Affecting the Usability and Sustainability of Wearable Devices for Health Management - A Review of User Experience from a Convergence Perspective of Technology, Psychology and Interaction
Authorship: Third author · Organizer: Korean Institute of Information Scientists and Engineers, Communications of the KIISE · Pages: Vol. 32, No. 11 (Issue 306), p37-45 (Korean)

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